

ALTUS AFB, OK, USA

WMO: 723520

Lat: 34.650N

Lon: 99.267W

Elev: 1382

StdP: 13.98

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13902

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.9	19.4	1.4	6.2	25.8	6.1	7.9	28.3	31.5	41.4	28.3	42.8	9.1	20	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.1	104.6	72.9	101.8	73.0	99.1	73.1	78.0	93.7	76.8	92.2	75.6	90.6	12.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.2	134.4	82.0	72.9	128.6	81.4	72.0	124.6	81.0	42.4	93.6	41.1	92.0	40.0	90.4	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.7	24.9	22.0	DB	7.6	107.7	4.8	3.8	4.1	110.4	1.3	112.6	-1.4	114.7	-4.9	117.5	(4)
(5)			WB	6.3	79.4	4.4	1.9	3.2	80.8	0.6	81.9	-1.9	83.0	-5.0	84.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.8	40.3	45.0	53.4	61.1	71.4	80.8	84.2	83.4	75.6	63.6	51.5	41.5		
	DBStd	17.68	9.54	10.69	10.23	8.31	7.86	5.79	4.89	4.78	6.99	8.76	8.86	9.00		
	HDD50	1001	322	201	82	13	0	0	0	0	0	10	84	289		
	HDD65	3181	766	563	375	166	31	0	0	0	10	132	409	729		
	CDD50	5661	22	60	189	348	665	925	1061	1035	769	431	131	25		
	CDD65	2364	0	2	17	50	231	475	596	570	329	89	5	0		
	CDH74	29731	20	77	292	807	2647	5969	8102	7176	3475	1033	120	13		
(13)	CDH80	16374	2	18	81	294	1242	3375	4936	4255	1777	371	21	2		
(14)	Wind	WSAvg	10.4	10.0	10.9	12.1	12.7	11.5	9.2	8.7	9.1	9.9	9.9	9.5		
(15) Precipitation	PrecAvg	27.1	0.9	1.2	1.7	2.1	4.1	4.0	2.0	2.8	3.1	2.6	1.5	1.1		
	PrecMax	42.4	4.1	4.5	4.9	7.4	13.3	11.0	7.5	7.4	10.0	11.8	7.2	4.8		
	PrecMin	10.4	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	7.3	1.0	1.1	1.3	1.6	2.9	2.6	1.7	2.2	2.4	2.6	1.6	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.0	82.0	87.8	93.4	100.8	108.0	108.2	107.9	101.9	93.0	83.1	74.8		
		MCWB	55.3	58.2	61.7	62.7	69.0	72.3	72.3	72.2	71.1	67.6	63.0	56.0		
	2%	DB	69.3	75.2	81.4	87.6	95.8	102.3	104.7	103.8	97.7	88.3	77.5	67.2		
		MCWB	51.2	55.9	61.2	63.2	69.4	72.5	73.6	73.3	71.5	67.2	60.3	51.8		
	5%	DB	63.8	69.7	76.7	82.5	91.4	98.6	101.8	100.4	94.2	84.1	72.7	61.7		
		MCWB	49.5	53.1	59.8	62.3	69.1	72.7	73.6	73.7	71.1	65.8	57.6	50.9		
	10%	DB	58.3	63.6	71.7	78.3	87.0	95.0	98.9	97.6	90.4	80.1	67.4	56.8		
		MCWB	46.2	50.9	56.8	61.4	68.5	72.7	73.6	73.4	70.2	63.7	55.2	47.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.5	63.7	68.4	71.0	75.6	79.7	80.0	78.9	77.1	73.5	66.0	61.7		
		MCDB	68.6	70.2	78.1	81.1	88.0	94.9	95.2	95.0	91.5	84.9	77.0	66.1		
	2%	WB	53.9	59.1	65.8	68.6	73.8	77.5	77.9	77.4	75.1	71.2	62.9	56.6		
		MCDB	64.4	69.0	75.1	78.3	87.1	92.1	94.2	93.8	88.3	80.4	71.9	62.0		
	5%	WB	50.1	55.5	62.9	66.6	72.5	76.1	76.6	76.1	73.7	69.1	60.4	52.2		
		MCDB	61.4	66.9	71.0	76.0	85.1	90.4	92.6	92.1	86.7	78.0	69.4	60.7		
	10%	WB	46.8	51.9	59.5	64.2	70.9	74.8	75.4	75.1	72.6	66.9	57.4	47.9		
		MCDB	57.7	63.6	68.9	73.3	82.6	88.9	90.9	90.6	84.7	76.7	65.2	55.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	26.8	26.5	26.6	26.8	24.4	24.0	25.1	24.9	23.9	26.2	25.9	23.6		
		MCDBR	37.4	37.2	34.0	35.2	30.6	30.0	30.4	29.9	29.6	31.0	33.2	32.8		
		MCWBR	21.8	20.7	17.4	16.3	10.8	8.9	7.9	8.4	9.4	13.4	17.9	19.7		
		MCDBR	34.6	33.0	26.8	26.5	24.7	23.9	24.8	24.6	23.5	23.6	27.7	28.4		
	5% WB	MCWBR	21.2	20.2	15.3	15.3	11.0	8.9	8.2	8.4	8.5	12.4	17.2	19.3		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.298	0.331	0.365	0.387	0.414	0.416	0.407	0.375	0.343	0.311	0.285		
	taud		2.535	2.497	2.423	2.353	2.362	2.323	2.357	2.329	2.394	2.471	2.490	2.529		
	Ebn at Noon		297	303	299	292	285	276	275	276	280	282	282	289		
	Edh at Noon		25	30	35	39	39	41	39	40	35	30	26	24		
(44) All-Sky Solar Radiation	RadAvg		964	1158	1542	1877	2026	2231	2224	2023	1688	1297	1004	819		
	RadStd		95	154	98	145	210	186	125	117	113	173	124	82		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46) N/A
(47)	Regional (25 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47) N/A

Nomenclature: See separate page